1ICPC317

AY 2024-25

Ubiquitous Sensing, Computing & Communication Laboratory

Quality Laboratory Manual

<u>Experiment No. 01</u>

To Understand the Phases in a Software Development Project.

Course Instructor – **Mr. Samish N Kamble**
ASSISTANT PROFESSOR

Experiment No. 01

Title of Experiment: To Understand Raspberry pi 4 model B board.

Aim of Experiment: To understand the details of placement and specification of different components on the raspberry pi 4 model B board and installation of Raspbian operating system on the board.

System Requirements – Raspberry pi 4 Board Win 10 and above OS, 4GB RAM, 2.33 GHz Processor

Software/s Needed for Experiment –

Experiment Objectives:

- To Identify and Understand the Components on the Raspberry Pi 4 Model B Board.
- To Demonstrate and Apply the Process of Installing Raspbian Operating System on the Raspberry Pi 4 Model B.
- To Configure and Evaluate the Raspberry Pi 4 for Basic Use.

Experiment Outcomes:

- **Demonstrate understanding of the Raspberry Pi 4 Model B hardware components** and their functions.
- **Install and configure the Raspbian operating system** on the Raspberry Pi 4 Model
- **Evaluate the functionality** of the Raspberry Pi 4 by verifying proper operation of peripherals and network connections.
- Gain practical experience in **hardware identification, OS installation, and system configuration.**
- Develop skills for using single-board computers for **various applications**.

Theory:

The Raspberry Pi 4 Model B is the latest version of the low-cost Raspberry Pi computer. The Pi isn't like your typical device; in its cheapest form it doesn't have a case, and is simply a credit-card sized electronic board -- of the type you might find inside a PC or laptop, but much smaller.

Raspberry Pi 4 Model B features a high-performance 64-bit quad-core processor, dual-display support at resolutions up to 4K via a pair of micro HDMI ports, hardware video decode at up to 4Kp60, up to 8GB of RAM, dual-band 2.4/5.0 GHz wireless LAN, Bluetooth 5.0, Gigabit Ethernet, USB 3.0, and PoE capability (via a separate PoE HAT add-on). For the end user, Raspberry Pi 4 Model B provides desktop performance comparable to entry-level x86 PC systems. This product retains backwards compatibility with the prior-generation Raspberry Pi 3 Model B+ and has similar power consumption, while offering substantial increases in processor speed, multimedia performance, memory, and

connectivity. The dual-band wireless LAN and Bluetooth have modular compliance certification, allowing the board to be designed into end products with significantly reduced compliance testing, improving both cost and time to market.

Specification

Processor: Broadcom BCM2711, quad-core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5GHz

Memory: 1GB, 2GB, 4GB or 8GB LPDDR4 (depending on model) with on-die ECC

Connectivity: 2.4 GHz and 5.0 GHz IEEE 802.11b/g/n/ac wireless LAN,

Bluetooth 5.0, BLE • Gigabit Ethernet

USB 3.0 ports • 2 × USB 2.0 ports. GPIO:

Standard 40-pin GPIO header (fully backwards-compatible with previous boards)

Video & sound: • 2 × micro HDMI ports (up to 4Kp60 supported) •

2-lane MIPI DSI display port

2-lane MIPI CSI camera port

4-pole stereo audio and composite video port Multimedia: H.265 (4Kp60 decode); H.264 (1080p60 decode, 1080p30 encode);

OpenGL ES, 3.0 graphics SD card support: Micro SD card slot for loading operating system and data storage

Input power: • 5V DC via USB-C connector (minimum 3A1) • 5V DC via GPIO header (minimum 3A1)

Power over Ethernet (PoE)–enabled (requires separate PoE HAT)

Environment: Operating temperature 0–50ºC Production lifetime:

Raspberry Pi 4 Model B will remain in production until at least January 2034.

Compliance: For a full list of local and regional product approvals, please visit pip.raspberrypi.com.

Operating System for Raspberry Pi 4

The Raspberry Pi 4 supports various operating systems, including:

1. Raspberry Pi OS (Official OS)

- A Debian-based Linux distribution optimized for Raspberry Pi.
- Comes in **three variants**:
  - **Raspberry Pi OS Full** (with desktop and recommended software)
  - **Raspberry Pi OS Desktop** (without extra software)
  - **Raspberry Pi OS Lite** (minimal, command-line only)
- Features:
  - Pre-installed Python, Scratch, and other development tools.
  - Lightweight and optimized for Raspberry Pi hardware.
  - Supports GPIO access for embedded applications.

2. Ubuntu

- A popular Linux distribution, available in **Server** and **Desktop** versions.
- Provides better support for advanced development and cloud-based applications.

3. Other Linux-based OS

- **Kali Linux** (for penetration testing and security analysis).
- **LibreELEC** (for media center applications).
- **RetroPie** (for gaming emulation).

4. Non-Linux OS

- **Windows 10 IoT Core** (for IoT and industrial applications).

- **RISC OS** (lightweight OS for experimentation).

Installation of Raspberry Pi OS

1. Downloading and Preparing the OS

- Download **Raspberry Pi OS** from the official Raspberry Pi website.
- Use **Raspberry Pi Imager** or software like **Balena Etcher** to flash the OS onto a microSD card.

2. Booting the Raspberry Pi

- Insert the microSD card into the Raspberry Pi 4.
- Connect a keyboard, mouse, and display.
- Power on the Raspberry Pi using a USB-C power adapter.

3. Initial Setup

- Follow on-screen instructions to set up language, Wi-Fi, and updates.
- Enable SSH, VNC, or GPIO interface if needed.

Observations

1. **Successful Booting** – The Raspberry Pi 4 booted correctly, and the OS loaded with a functional GUI or terminal.
2. **System Performance** – The OS responded well to commands, with stable CPU and RAM usage.
3. **Board Description** – The Raspberry Pi 4 features a **Broadcom BCM2711** processor, **1GB/2GB/4GB/8GB RAM**, **40 GPIO pins**, **USB 3.0 ports**, and **dual micro HDMI outputs**, enabling versatile applications.

Conclusion:

The experiment successfully demonstrated the installation and operation of the Raspberry Pi 4's operating system. The board booted correctly, responded efficiently to commands, and supported various peripherals. The study also highlighted the hardware specifications of the Raspberry Pi 4, making it a powerful and versatile platform for embedded applications, IoT, and computing projects.

Expected Oral Questions:

1. What are the key hardware features of the Raspberry Pi 4 board?
2. Which operating systems can be installed on the Raspberry Pi 4?
3. What are the steps to install Raspberry Pi OS on a microSD card?
4. How can you check system performance on Raspberry Pi using terminal commands?
5. What are the common applications of Raspberry Pi in embedded systems and IoT?

FAQs in Interview:

1. What are the key differences between Raspberry Pi 4 and its previous versions?
2. Which operating system is officially recommended for Raspberry Pi, and why?
3. How do you install and boot an OS on Raspberry Pi 4?
4. What are the key commands used to check system performance on Raspberry Pi OS?
5. Explain the role of the microSD card in Raspberry Pi 4 and its ideal specifications.